DUT: ATmega16U2

File: dut.kicad_sch

Pi to DUT: level translators, reset, HWB

File: pi_io.kicad_sch

Clock override (option CLK) and Si5351 (option SI)

File: clock.kicad_sch

DUT power switch and power meter

File: dut_power.kicad_sch

High-voltage programming (option HV)

File: hv.kicad_sch

USB path switch

File: usb.kicad_sch

HAT: 40-pin header, ID EEPROM, USB-C loopback, LED

File: hat_io.kicad_sch

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TomSawyerLabs mcu-hil

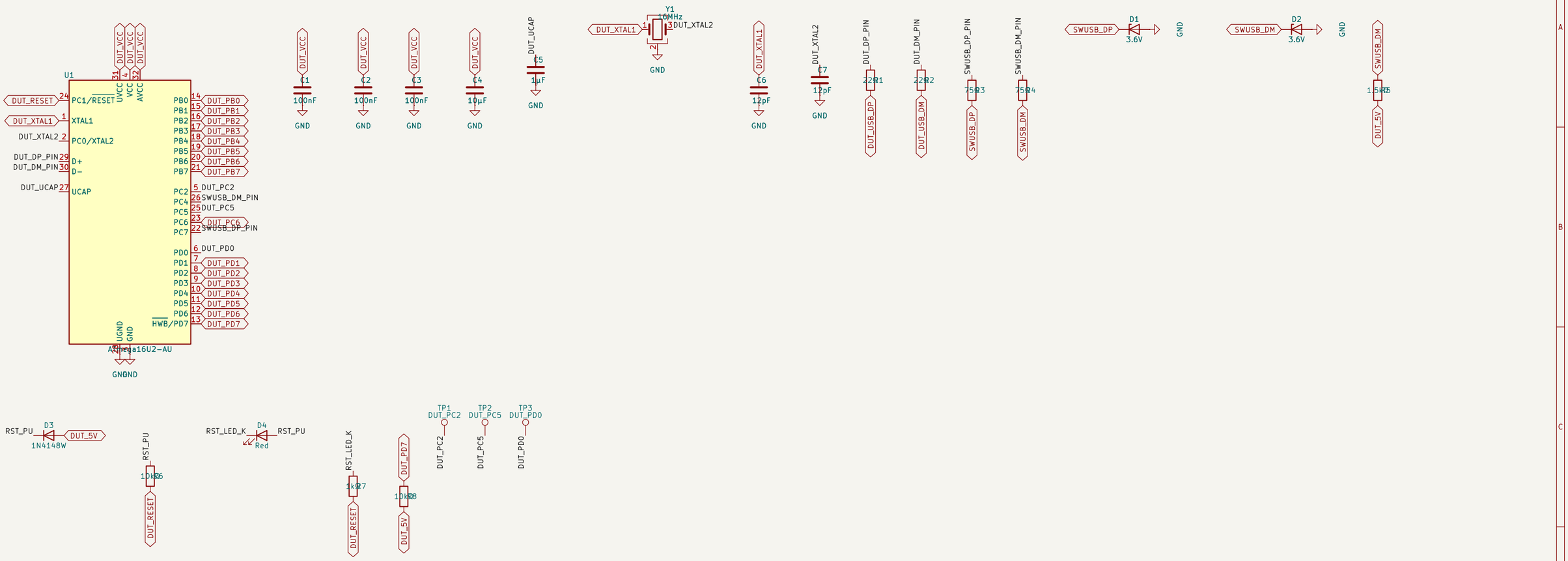
Sheet: /

File: mcu-hil-hat.kicad_sch

Title: mcu-hil v2: Pi HAT, ATmega16U2

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/8

ATmega16U2 at 5 V, 16 MHz. DUT_VCC is measured by the power meter; DUT_5V (before the shunt) feeds pull-ups.
Software USB (V-USB): D+ = PC7 (INT4), D− = PC4. 75R series and 3.6 V zeners keep the lines at USB levels.
Reset: the Pi side holds RESET low by default (pi_io sheet). The diode keeps 12 V (HVPP) off DUT_5V.



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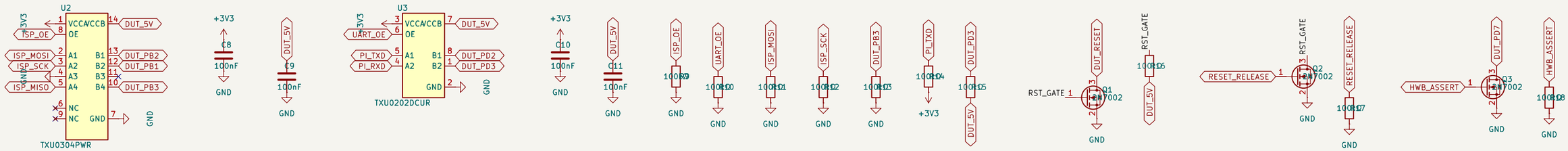
Sheet: /DUT: ATmega16U2/

File: dut.kicad_sch

Title: DUT: ATmega16U2

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 2/8

Translators: A side on +3V3 (Pi), B side on DUT_5V, so they go Hi-Z (off) whenever the DUT is off.
Every enable defaults off through an external pull-down (docs/v2-pinmap.md).
RESET: Q2 off (GPIO12 low, default) lets RST_GATE rise with DUT_5V, so Q1 holds RESET low.



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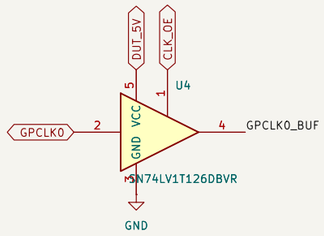
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Sheet: /Pi to DUT: level translators, reset, HWB/
File: pi_io.kicad_sch

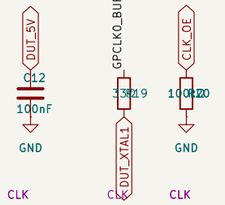
Title: Pi to DUT: level translators, reset, HWB

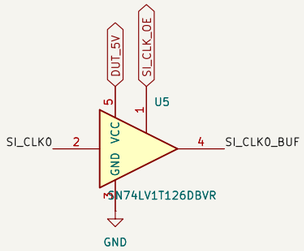
Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/8

Two tri-state buffers drive XTAL1 over the crystal: GPCLK0 (GPIO4, enable GPIO18) or Si5351 CLK0 (enable GPIO27). Both default off. Software must never enable both. HVPP pulses XTAL1 through U(GPCLK0).

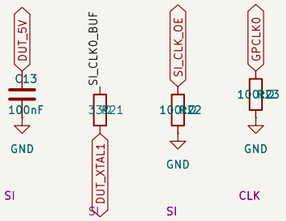


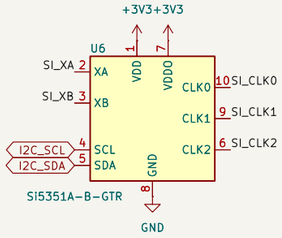
CLK



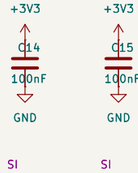


SI

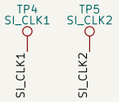




SI







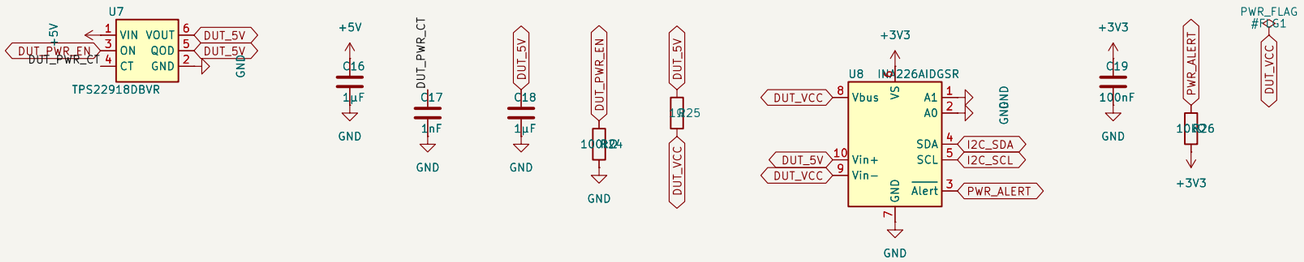
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Sheet: /Clock override (option CLK) and Si5351 (option SI)/
File: clock.kicad_sch

Title: Clock override (option CLK) and Si5351 (option SI)

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 4/8

TPS22918: off by default (GPIO22 pull-down), set slew (CT), quick output discharge (QOD tied to VOUT).
INA226 measures DUT_VCC (the MCU only) across the 1 ohm shunt: 81.9 mA full scale, 2.5 uA steps.
INA228 (20-bit, same footprint and pinout) is a drop-in upgrade when JLC has stock: 164 mA, 0.3 uA steps.



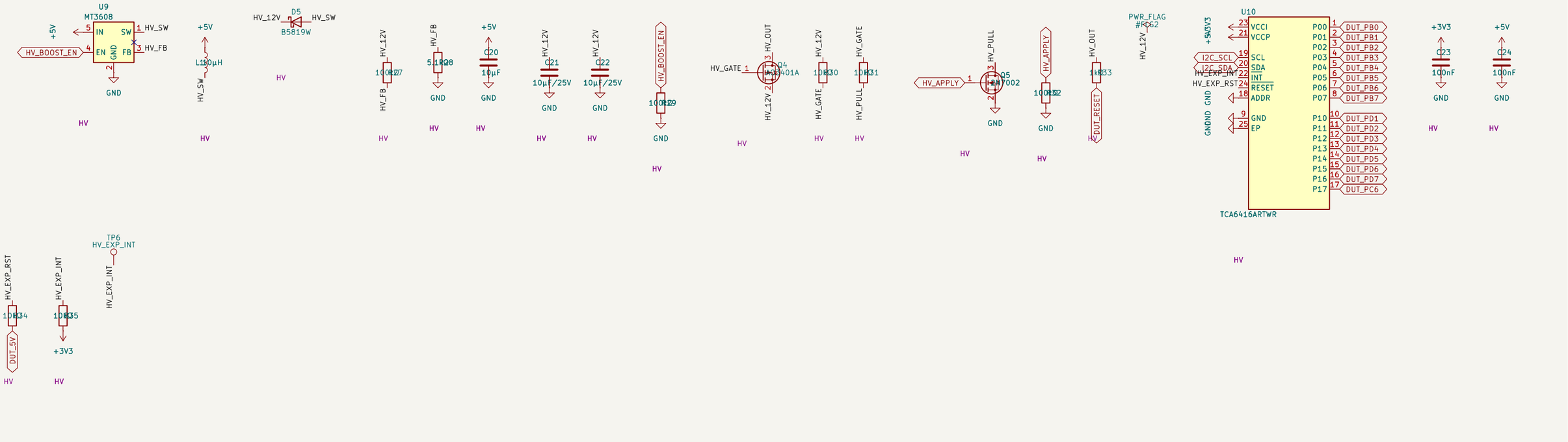
Generated by tools/schgen. Edit the generator, not this file.
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Sheet: /DUT power switch and power meter/
File: dut_power.kicad_sch

Title: DUT power switch and power meter

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 5/8

12 V boost (MT3608; GPIO26) and a switch that puts it on RESET (GPIO25). Both default off.
FB divider 100k/5.1k: 12.36 V nominal. The 1k series resistor drops about 0.25 V at the 250 uA RESET current, so RESET sees 11.9–12.4 V (spec 11.5–12.5 V).
TCA6416A: VCCP on +5V (always up before +3V3, as TI recommends). ~RESET is pulled up to DUT_5V, so the expander's ports are held as inputs (Hi-Z) whenever the DUT is off. I2C address 0x20.
HVPP map (16U2 table 25-9): P0 = PB0-7 data; P10 PD1 RDY/BSY, P11 PD2 OE, P12 PD3 WR, P13 PD4 BS1, P14 PD5 XA0, P15 PD6 XA1, P16 PD7 PAGEL, P17 PC6 BS2.



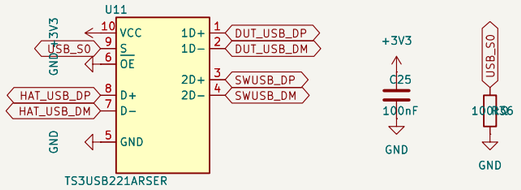
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TomSawyerLabs mcu-hil

Sheet: /High-voltage programming (option HV)/
File: hv.kicad_sch

Title: High-voltage programming (option HV)

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 6/8

USB_S0 (GPIO20) low: the loopback USB-C port to the 16U2 hardware USB. High: to software USB.



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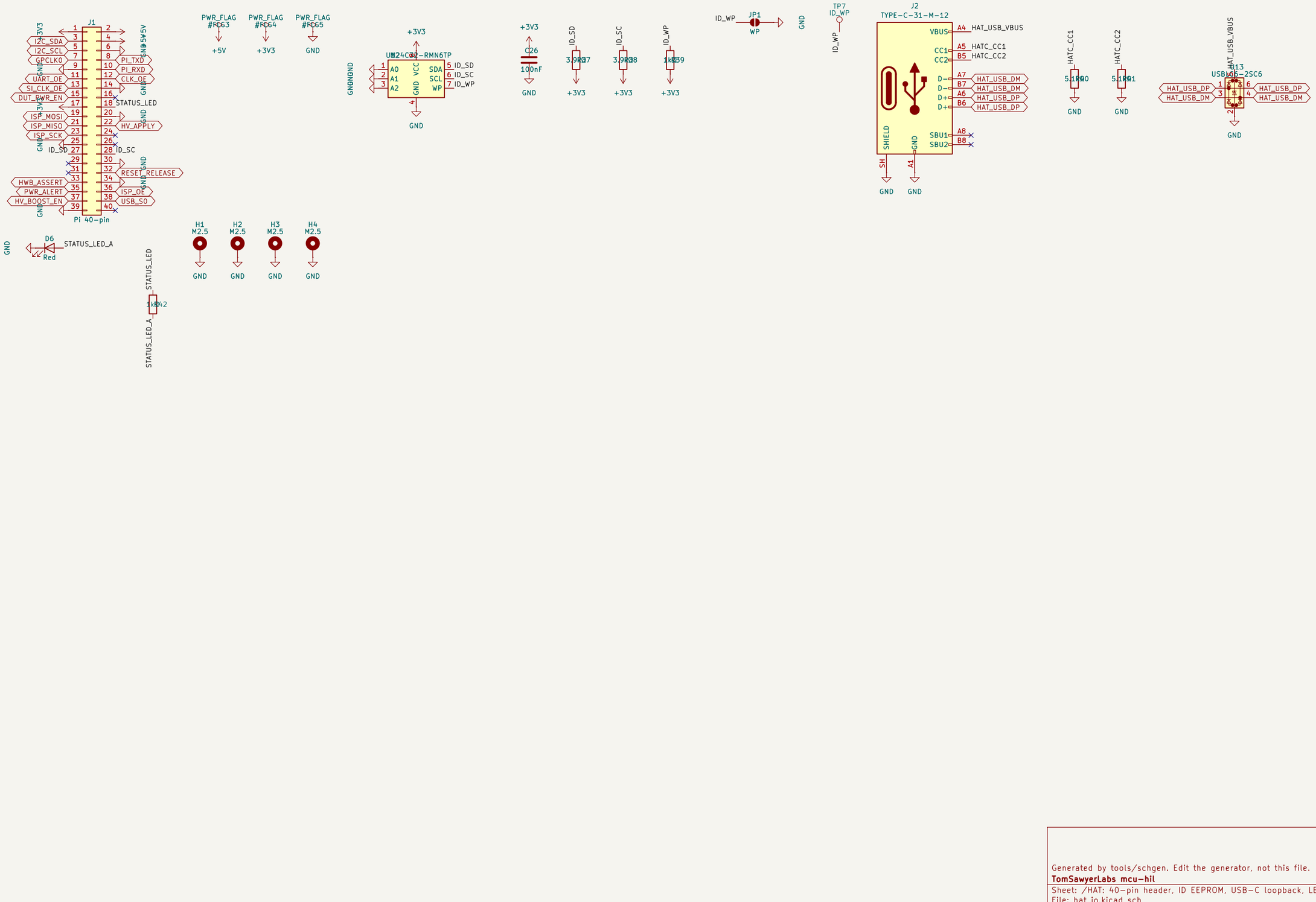
Sheet: /USB path switch/

File: usb.kicad_sch

Title: USB path switch

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 7/8

HAT+ rules: ID_SD/ID_SC carry only the EEPROM and 3.9k pull-ups to 3.3 V. EEPROM WP has a 1k pull-up; bridge JP1 (or drive the test point low) to write it. The HAT never powers the Pi.
USB-C here is a data port only: a short cable back to one of the Pi's USB host ports. Its VBUS is not used.
GPIO2/3 already have ~2k pull-ups on the Pi. STANDBY (5 V up, 3.3 V down) is safe: every enable defaults off.



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Sheet: /HAT: 40-pin header, ID EEPROM, USB-C loopback, LED/
File: hat_io.kicad_sch

Title: HAT: 40-pin header, ID EEPROM, USB-C loopback, LED

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/8